

Signal Analysis & Telemetry Extraction Report

Target Recording: DEMO_SAT_QPSK_DEGRADED.iq | Format: IQ | Sample Rate: 500.0 kSps

SIGNAL HEALTH
46/100
STATUS: CRITICAL

ANALYST PRIORITY
HIGH
Critical parameter
anomal...

CONFIDENCE
99.0%
DSP DATA QUALITY

ANOMALIES
3
DETECTED ISSUES

1. Explainable Analysis & Executive Summary

Automated analysis completed for 'DEMO_SAT_QPSK_DEGRADED.iq'.

The recording contains 1 detected active segment(s) across a total duration of 1.000s. Spectral analysis identifies a center frequency at 100.985 kHz with an occupied bandwidth of 128.418 kHz and an SNR of 35.0 dB (Modulation profile: Frequency Shift Keying (FSK / 2-FSK/4-FSK)).

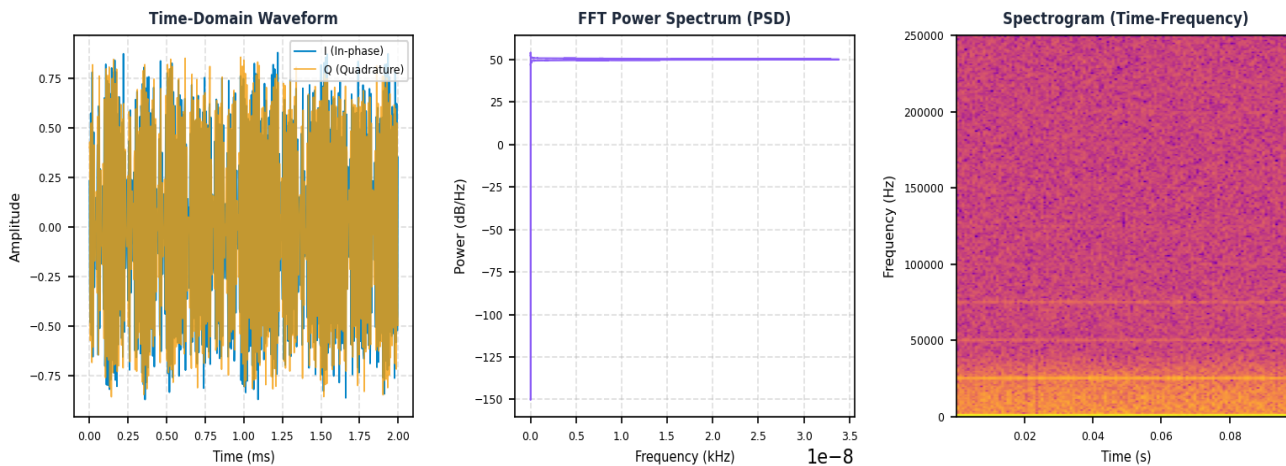
Benchmarked against historical baseline 'SAT-QPSK-REF-BASELINE'. Statistical drift analysis indicates baseline center frequency was 80037.0 Hz (observed: 100985.1 Hz) and nominal SNR was 48.2 dB.

Detected 3 anomaly condition(s):

- Center Frequency (CRITICAL): Center frequency drifted by +26.2% (20948.2 Hz) from baseline mean (13.1 σ deviation). Possible oscillator frequency drift or Doppler shift.
- Signal-to-Noise Ratio (SNR) (CRITICAL): SNR degraded by 13.2 dB below historical baseline. Indicates elevated channel noise floor, attenuation, or RF interference.
- Occupied Bandwidth (HIGH): Occupied bandwidth expanded/compressed by +289.6% from nominal baseline. Indicates symbol rate change, non-linear distortion, or spectral regrowth.

Overall Signal Health Score is 46.0/100 (CRITICAL), assigning this file to HIGH PRIORITY for engineer triage.

2. Time-Domain, Spectral & Time-Frequency Visualization



3. Extracted Signal Parameters

MEASURED PARAMETER	VALUE	DERIVED / ESTIMATED PARAMETER	VALUE
Peak Frequency	99.854 kHz	Occupied Bandwidth (99% OBW)	128.418 kHz
Spectral Centroid	100.985 kHz	-3 dB Bandwidth	15.625 kHz

Signal Power (dBFS)	-3.59 dB	-10 dB Bandwidth	28.076 kHz
Est. Noise Floor	-80.64 dB	Signal Stability Index	0.913 / 1.0
Measured SNR	34.98 dB	Crest Factor / PAPR	1.512 (3.59 dB)
Peak Amplitude	1.0 V	Spectral Flatness (Entropy)	0.0217
RMS Amplitude	0.6615 V	Modulation Classifier Hint	Frequency Shift Keying (FSK / 2-FSK/4-FSK) (88.0%)

4. Signal Fingerprint & 'What Changed?' Analysis (Fingerprint ID: FP-E4D52EF5A865D8B9)

PARAMETER	BASELINE MEAN	CURRENT VALUE	DELTA (%)	DIR	STATUS
Center Frequency	80036.97 Hz	100985.13 Hz	+26.2%	↑	Anomalous High
Bandwidth (99% OBW)	32958.98 Hz	128417.97 Hz	+289.6%	↑	Anomalous High
Signal-to-Noise Ratio (SNR)	48.22 dB	34.98 dB	-13.2%	↓	Anomalous Low
Signal Power	-2.27 dBFS	-3.59 dBFS	-1.3%	→	Nominal
Signal Stability	0.913 index	0.913 index	+0.0%	→	Nominal
RMS Amplitude	0.77 V	0.661 V	-14.0%	↓	Anomalous Low

5. Anomaly Investigation Log (3 events)

SEVERITY	AFFECTED PARAMETER	OBSERVED VALUE	ENGINEERING REASON
CRITICAL	Center Frequency	100985.1 Hz	Center frequency drifted by +26.2% (20948.2 Hz) from baseline mean (13.1σ deviation). Possible oscillator frequency drift or Doppler shift.
CRITICAL	Signal-to-Noise Ratio (SNR)	35.0 dB	SNR degraded by 13.2 dB below historical baseline. Indicates elevated channel noise floor, attenuation, or RF interference.
HIGH	Occupied Bandwidth	128418.0 Hz	Occupied bandwidth expanded/compressed by +289.6% from nominal baseline. Indicates symbol rate change, non-linear distortion, or spectral regrowth.